Department of Artificial Intelligence and Machine Learning

Date: 15/04/2026	Test – I	Max. Marks : 60
Semester: VI	UG	Duration : 1.5 hrs
Course Title: Natural Language Processing and Transformers		Course Code: AI363IA

Part – A

Q. No	Question	M	BT	CO
1.1	_____ and _____ are data acquisition techniques in NLP.	2	1	1
1.2	How is raw text prepared to ensure consistency before applying NLP techniques?	2	1	1
1.3	With an example differentiate Code mixing and transliteration .	2	1	2
1.4	Analyse the different similarity measures with reference to accuracy and give example	2	2	2
1.5	Give the difference between stemming and lemmatization with an example.	2	2	2

Part – B

Q. No	Question	M	BT	CO
1	a. Elaborate Heuristic based NLP b. "Deep Learning is not the silver bullet for NLP". Justify the given phrase	6+4	2	1
2	With a neat diagram, elaborate the NLP pipe line in detail.	10	1	1
3	a. What is Word sense disambiguation? Consider the example "apple store near me". Implement Lesk algorithm to detect disambiguation for the above example. b. An NLP model performs well on training data but fails in real-world applications with informal text, slang, and ambiguous expressions. Identify and explain the key limitations causing this performance gap.	6+4	3,2	2,1
4	A university digital library provides a search feature for students to find study materials. However, students often use different words for the same concept, causing the system to miss relevant results. (a) Design a system that expands the user query using WordNet relations, including Synonyms, Hypernyms and Hyponyms. (You can create sample document database for the same) (b) Give the advantage of using WordNet for the system	8+2	3	2
5	A job portal allows users to search for job postings using keywords. A user enters the query "developing applications", but relevant job listings contain words like "developer", "developed", "application", "apps", etc. List and apply different types of stemmers to the words. Identify and justify the best suited stemmer in this scenario.	10	3,4	2